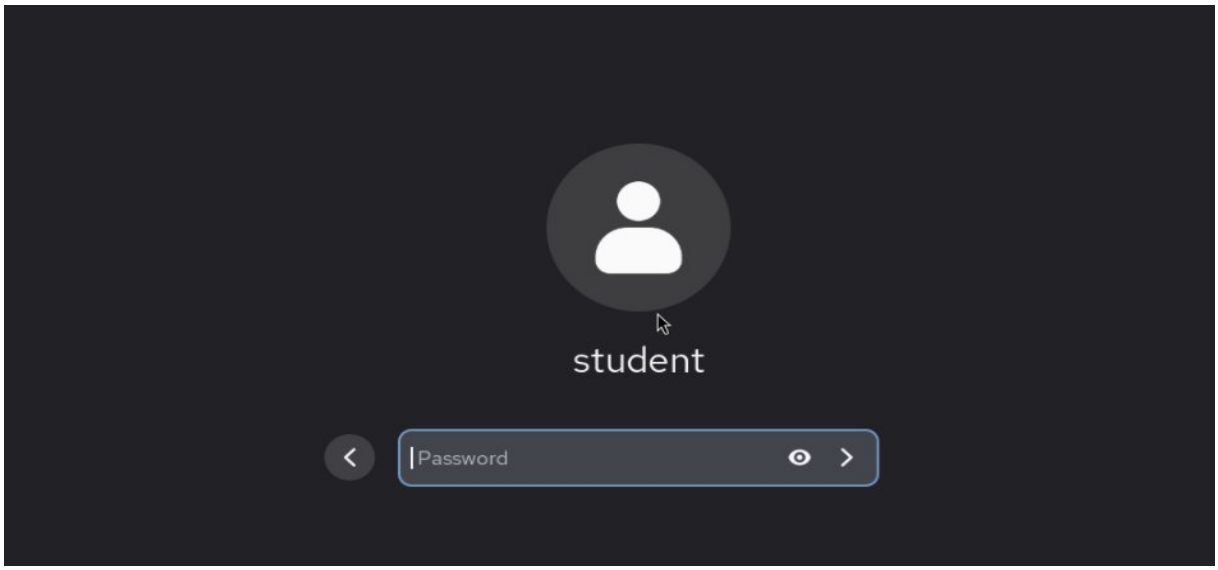
Lab Number : 29 Date:2026/03/15

Title: Users , Group , Permissions

1.Create a user named “students”:



2. Login from “student” user then create files and folders according to following tree structure [d1 = directory & f = file:

```
student@localhost:~$ tree /home/student/d1
/home/student/d1
├── d2
│   ├── d5
│   │   ├── f2
│   │   └── f3
│   └── d6
│       ├── f4
│       └── f5
├── d3
│   └── d8
├── d4
│   └── d7
│       └── f6
└── f1

8 directories, 6 files
```

Commands:

```
student@localhost:~$ mkdir d1
student@localhost:~$ cd d1
student@localhost:~/d1$ mkdir d2 d3 d4
student@localhost:~/d1$ touch f1
student@localhost:~/d1$ mkdir -p d2/d5 d2/d6
student@localhost:~/d1$ touch d2/d5/f2 d2/d5/f3
student@localhost:~/d1$ touch d2/d5/f4 d2/d6/f5
student@localhost:~/d1$ mkdir d3/d8
student@localhost:~/d1$ mkdir d4/d7
student@localhost:~/d1$ touch d4/d7/f6
```

3. Change permission of file f1 so that owner will get full permission, group member will get read & execute permission &others will only get read permission:

Command:

```
student@localhost:~$ chmod 754 d1/f1
```

Output:

```
student@localhost:~$ ls /home/student/d1 -all
total 4
drwxr-xr-x.  5 student student   46 Mar 16 06:49 .
drwx------. 15 student student 4096 Mar 16 06:49 ..
drwxr-xr-x.  4 student student   26 Mar 16 06:50 d2
drwxrwxrwx.  3 student student   16 Mar 16 06:51 d3
drwxr-xr-x.  3 student student   16 Mar 16 06:51 d4
-rwxr-xr--.  1 student student    0 Mar 16 06:49 f1
```

4. Change permission of file f2 such that owner's & group members will get read & write permission but others will get no permission:

Command:

```
student@localhost:~$ chmod 660 d1/d2/d5/f2
```

Output:

```
student@localhost:~$ ls /home/student/d1/d2/d5  -all
total 0
drwxr-xr-x. 2 student student 26 Mar 16 07:01 .
drwxr-xr-x. 4 student student 26 Mar 16 06:50 ..
-rw-rw----. 1 student student  0 Mar 16 06:50 f2
-rw-r--r--. 1 student student  0 Mar 16 06:50 f3
```

5. Change permission & directory d3 such that all categories of works have full permission:

Command:

```
student@localhost:~$ chmod 777 d1/d3
```

Output:

```
student@localhost:~$ ls /home/student/d1/d3  -all
total 0
drwxrwxrwx. 3 student student 16 Mar 16 06:51 .
drwxr-xr-x. 5 student student 46 Mar 16 06:49 ..
drwxr-xr-x. 2 student student  6 Mar 16 06:51 d8
```